

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SCST1223
PROBABILITY AND STATISTICAL DATA ANALYSIS
Section : 04
Project 1: Common Study Habits Among UTM Students
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1.0 Introduction or Background

Common study habits are the regular practices that students implement to learn and retain knowledge effectively. These habits include setting up a study group and studying with disciplined friends, taking detailed notes during lectures and setting a consistent study schedule. Having good study habits is essential because it helps improve a student's attitude towards learning. The keys to better learning and better academic performance in schools are good teachers, good study environment, course of study, parents' cooperation, high quality books and, the most important, the study habits (Robinson, 2000).

However, in this age of entertainment, there are many things that can distract students' attention from studying, possibly leading to declined academic performance. Also, in recent years, students tend to underestimate the importance of good study habits.

Hence, this study aims to investigate the common study habits of Universiti Teknologi Malaysia (UTM) students. This study also analyzes the relationship between study habits and academic results.This research will provide beneficial insights about effective study habits that could help many students to improve their academic performance. From the data, it is expected to see a positive relationship between good study habits and higher academic performance, such as students with higher CGPA might prepare in advance or study more consistently.

This study focuses on UTM students as a target group. The survey is distributed in many social media platforms such as Whatsapp. The aim of the sample size in this study is 60 respondents, might be adequate for simple descriptive studies (Xu, 2011).

2.0 Data Collection

	Questions	Answers	Level of Measurement	Variable to be Measured
1	What is your gender?	Male / Female	Nominal	Gender
2	What is your age?	Below 18 / 18 - 24 / 25 - 34 / 35 - 44 / 45 - 54 / 55 - 64 / Above 65	Ordinal	Age
3	What is your CGPA?	Below 2.0 / 2.0 - 2.5 / 2.5 - 3.0 / 3.0 - 3.5 / Above 3.5	Ordinal	CGPA
4	What is your level of study?	Diploma / Degree / Master / Phd	Nominal	Level of study
5	What is your major of study?	Computing / Engineering / Artificial Intelligence / Management / Science / Social Science / Built Environment / others	Nominal	Major of study
6	What are your regularly used study techniques?	Group Study / Flashcards / Practice Questions / Taking Notes / Watching Online Tutorials	Nominal	Study techniques used regularly
7	Which of the studying environments do you prefer the most?	Library / Room / Cafe / Study Hall / Outdoors Others	Nominal	Preferred study environment
8	What is your average revision time per day? (hours)	1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16 / 17 / 18 / 19 / 20 / 21 / 22 / 23 / 24	Ratio	Average revision time per day
9	How many consecutive hours can you study before needing a break?	1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16 / 17 / 18 / 19 / 20 / 21 / 22 / 23 / 24	Ratio	Consecutive hours can students study

10	How long is your break before you start your next revision session?	1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16 / 17 / 18 / 19 / 20 / 21 / 22 / 23 / 24	Ratio	Length breaks that students need before continue studying
11	How many days in advance do you typically begin studying for exams?	1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / others	Ratio	Days in advance do students begin studying for exams
12	How do you feel about the effectiveness of your current study techniques?	1 / 2 / 3 / 4 / 5	Ordinal	Effectiveness of students' current study techniques

3.0 Data Analysis

3.1 Possible Analysis 1

Possible Analysis 1: Comparison of respondent gender, age (and others) simple descriptive analysis

In total, this survey has gained 64 respondents. The data from the survey is as below:

Part 1 - Personal Information

Part 2 - Questionnaire

Part 1: Personal Information

3.1.1 Gender Distribution

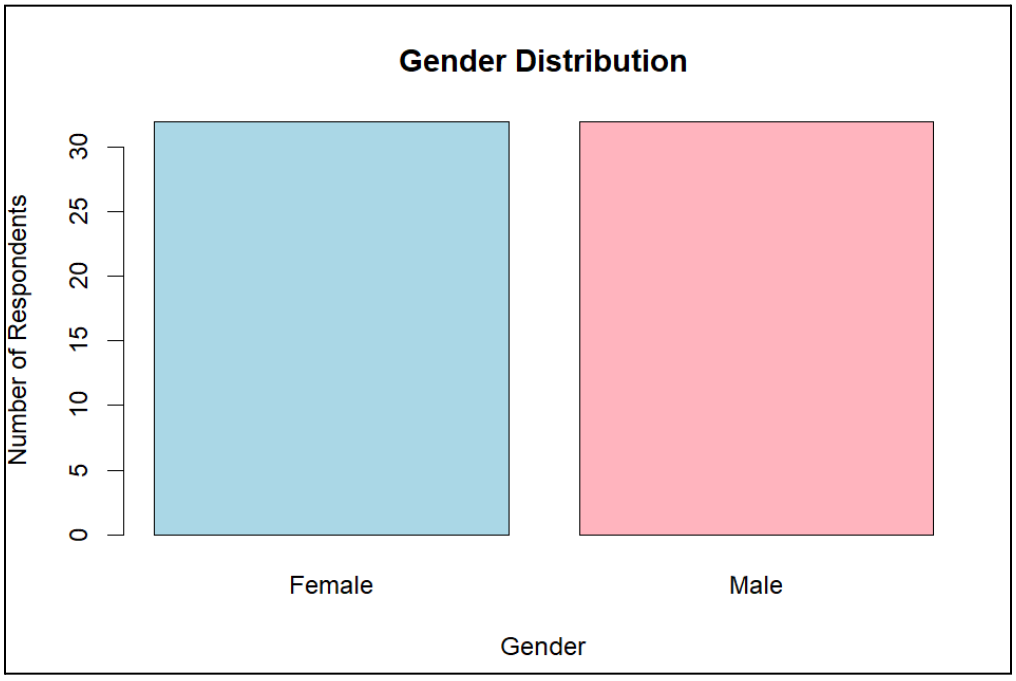


Figure 3.1 Bar chart of Gender Distribution

Based on the bar chart above, it is shown that there is an equal distribution between female and male respondents. There are approximately 30 respondents in each group. This also shows that this sample is beneficial because it is gender-balanced, which prevents the issues of potential gender bias in the analysis results and provides equal representation.

3.1.2 Age Distribution

Age	Frequency	Relative Frequency	Cumulative Frequency (CF)	Percent (%) of CF
Below 18	0	0	0	0
18 - 24	51	0.797	51	79.69
24 - 34	10	0.156	61	95.31
35 - 44	2	0.031	63	98.44
45 - 54	1	0.016	64	100.00
55 - 64	0	0	0	100.00
Above 65	0	0	0	100.00
Total	**64**	**1.000**	**64**	**100.00**

Table 3.1 Frequency Distribution Table of Age Distribution

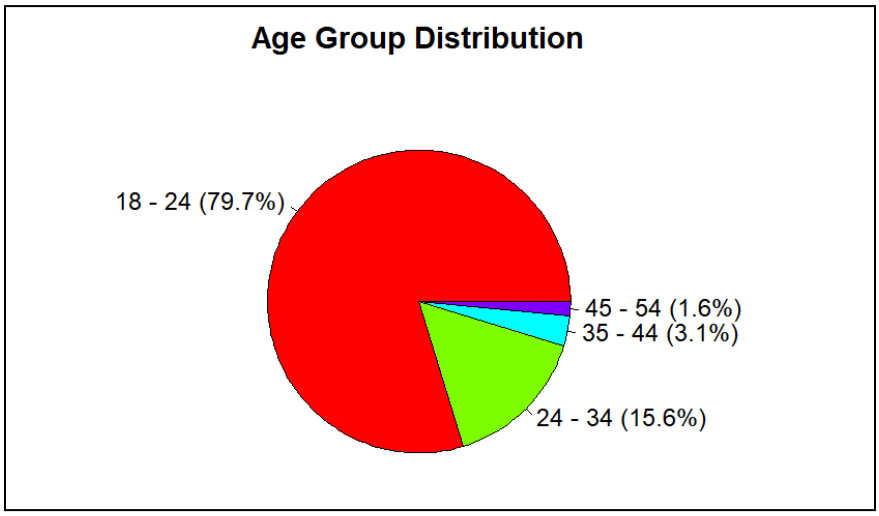


Figure 3.2 Pie chart of Age Distribution

According to the pie chart above, this survey is predominantly answered by young adults between the ages of 18 and 24, which makes up about 79.7% of the total 64 respondents. Other than this, the 24-34 age group forms the second largest segment at 15.6% of participants. Besides, there are few respondents from 35-44 and 45-54 age groups, which is 3.1% and 1.6% of the total respondents respectively. This shows that this analysis result is primarily about young adults.

3.1.3 CGPA Distribution

CGPA	Frequency	Relative Frequency	Cumulative Frequency (CF)	Percent (%) of CF
Below 2.0	0	0	0	0
2.0 - 2.5	0	0	0	0
2.5 - 3.0	7	0.109	7	10.94
3.0 - 3.5	21	0.328	28	43.75
Above 3.5	36	0.563	64	100.00
Total	**64**	**1.000**	**64**	**100.00**

Table 3.2 Frequency Distribution Table for CGPA Group Distribution

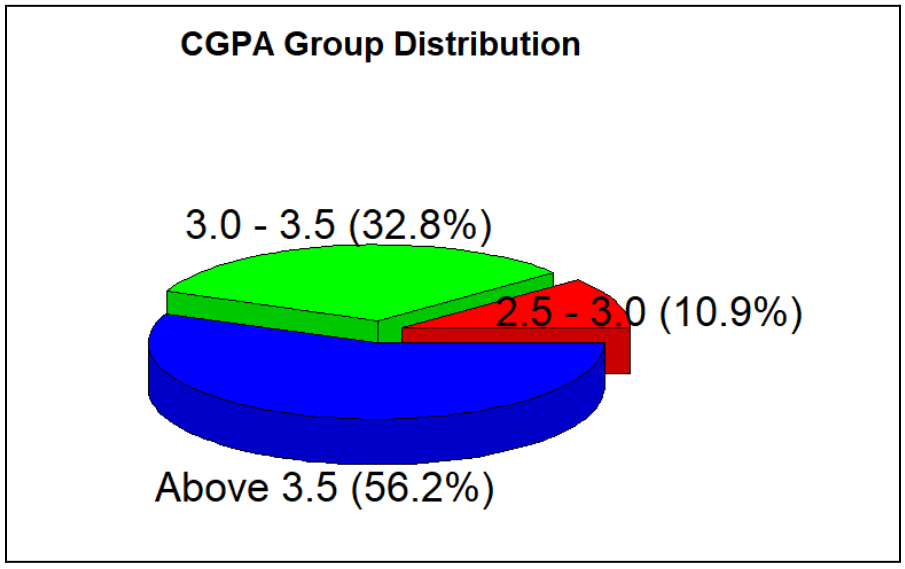


Figure 3.3 3D Pie Chart for CGPA Group Distribution

Based on the 3D pie chart above, most of the students in this survey have gained CGPA above 3.5, which makes up about 56.2% of the total 64 respondents. Other than this, students who have 3.0-3.5 CGPA forms the second largest segment at 32.8% of participants. Also, there are few respondents from the 2.5-3.0 CGPA range, which is 10.9% of the total respondents. This could show that this analysis result is considerably beneficial because it covers students who have different CGPA ranges.

3.1.4 Level of Study Distribution

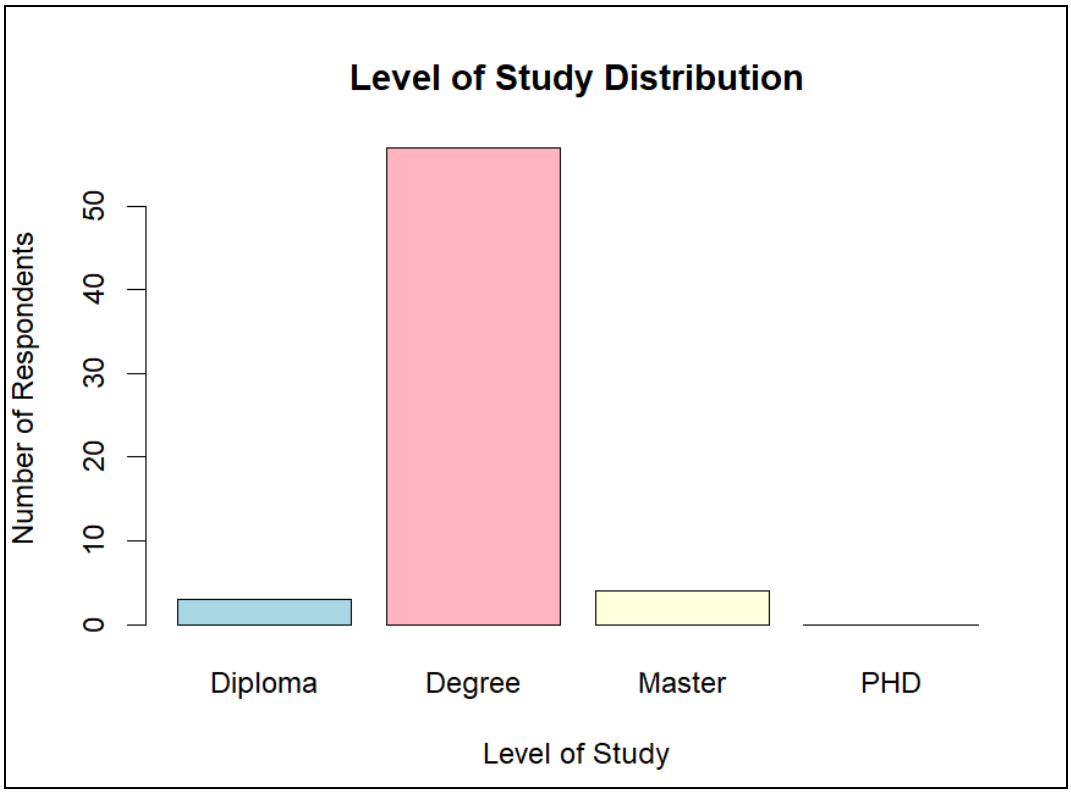


Figure 3.4 Bar Chart for Level of Study Distribution

Based on the bar chart above, it is shown that bachelor's degree students are the major respondents in this study. Bachelor's degree students in this study are around 55 out of 64 respondents. On the other hand, master's degree students and diploma holders are limited in number, with only 4 and 3 respondents respectively. In addition, there are no PHD students in the survey. This shows that this analysis result primarily reflects the perspectives of bachelor's degree students

3.1.5 Major of Study Distribution

Study Hours	Frequency	Relative Frequency	Cumulative Frequency (CF)	Percent (%) of CF
Computing	42	0.656	42	65.63
Engineering	14	0.219	56	87.5
Artificial Intelligence	5	0.078	61	95.31
Management	1	0.016	62	96.88
Science	1	0.016	63	98.44
Social Science	0	0	63	98.44
Built Environment	0	0	63	98.44
Others	1	0.16	64	100
Total	**64**	**1.000**	**64**	**100.00**

Table 3.3 Frequency Distribution Table for Major of Study Distribution

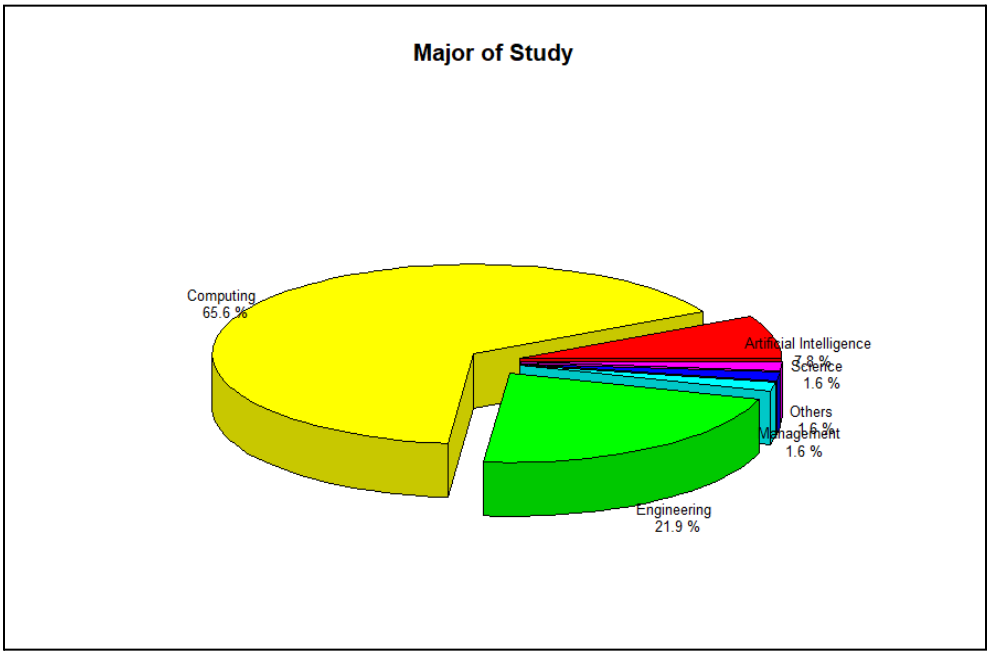


Figure 3.5 3D Pie Chart for Major of Study Distribution

According to the pie chart above, the majority of respondents are computing students, making up 65.6% of the total 64 respondents. Engineering students form the second largest group at 21.9%. Additionally, artificial intelligence students account for 7.8% of the participants. Meanwhile, management, science, and others each represent 1.6% of the respondents. There are no respondents from the social science and built environment fields. This shows that the analysis result primarily reflects the perspectives of computing students.

Part 2: Questionnaire

3.1.6 Regular Study Techniques Distribution

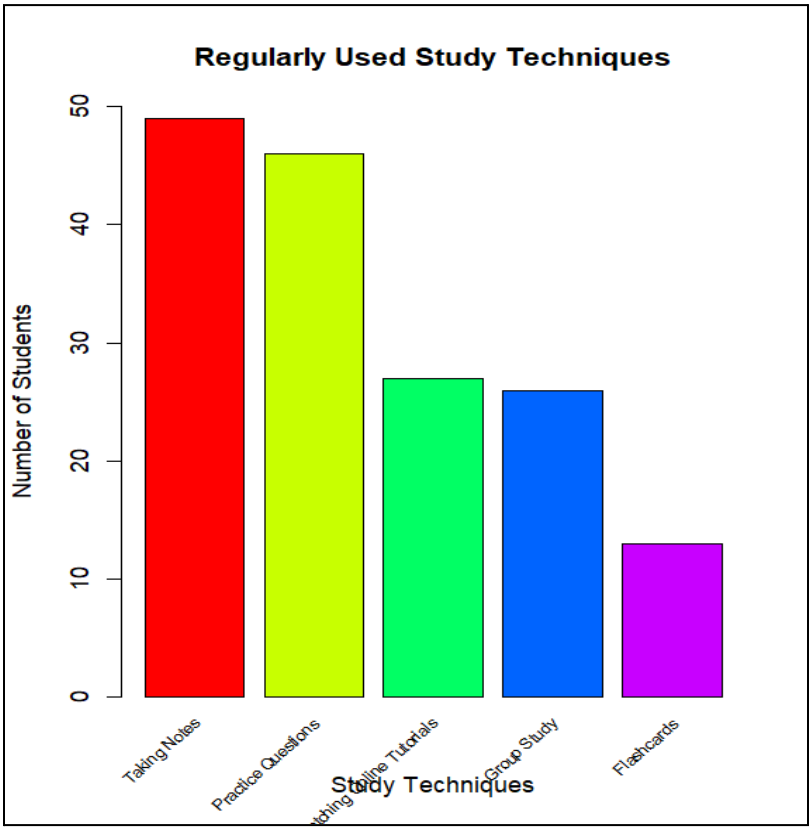


Figure 3.6 Bar Chart for Regular Study Techniques Distribution

Based on the bar chart above, it is shown that taking notes is the most commonly used study method among respondents, with 49 out of 64 students adopting this approach. practice questions are also highly utilized, with 46 students practicing this method. Meanwhile, watching online tutorials and group study are moderately popular, with 27 and 26 respondents, respectively. In contrast, Flashcards are the least used method, with only 13 students relying on them. This indicates that the analysis results mainly reflect a preference for note-taking and practicing questions among the respondents.

3.1.7 Preferred Environment Distribution

Study Hours	Frequency	Relative Frequency	Cumulative Frequency (CF)	Percent (%) of CF
Library	14	0.219	14	21.9
Room	30	0.469	44	46.9
Cafe	8	0.125	52	12.5
Study Hall	5	0.078	57	7.8
Outdoors	6	0.094	63	9.4
Others	1	0.016	64	1.6
Total	**64**	**1.000**	**64**	**100.00**

Table 3.4 Frequency Distribution Table for Preferred Environment Distribution

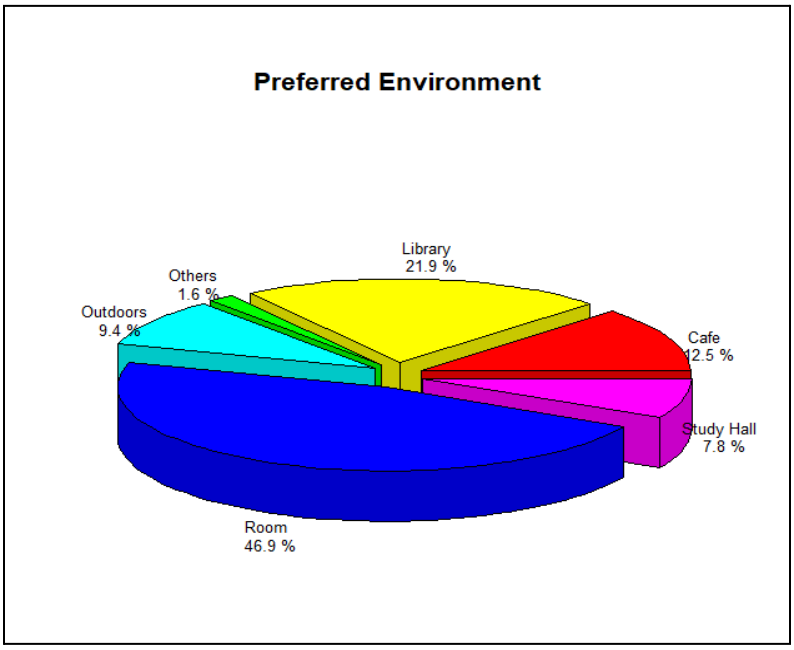


Figure 3.7 3D Pie Chart for Preferred Environment Distribution

According to the pie chart above, the majority of respondents prefer studying in their rooms, accounting for 46.9% of the total 64 respondents. The library is the second most preferred location, making up 21.9% of the responses. Other popular study locations include cafés at 12.5% and outdoor spaces at 9.4%. Meanwhile, study halls account for 7.8% of the preferences, and other locations represent only 1.6% of the total respondents. This indicates that the analysis result primarily reflects the study habits of students who prefer studying in their rooms.

3.1.8 Average Revision Per Day Distribution

Study Hours	Frequency
1	5
2	7
3	6
4	14
5	9
6	7
7	9
8	6
9	1
Total	**64**

Table 3.5 Frequency Distribution Table for Average Revision Per Day Distribution

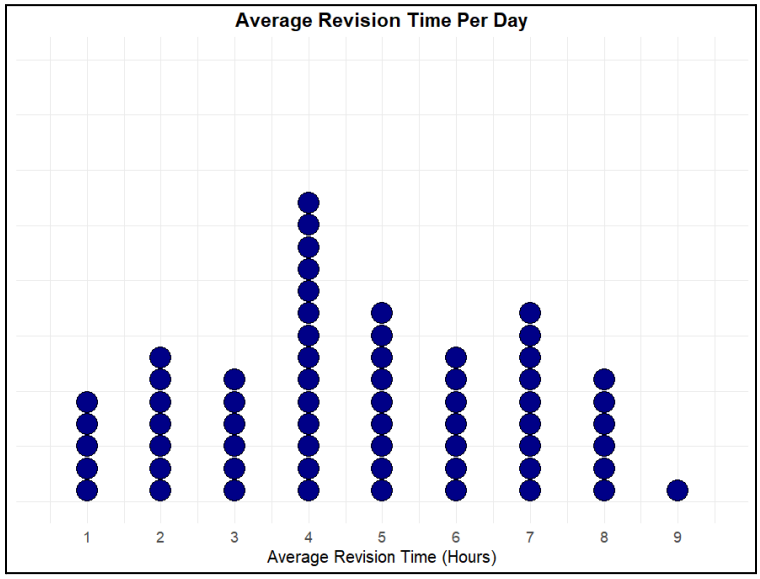


Figure 3.8 Dot Plot for Average Revision Per Day Distribution

According to the dot plot above, most respondents reported studying for 4 hours, representing the highest frequency with 14 out of 64 respondents. Study durations of 5 and 7 hours are also relatively common, with 9 respondents each. Study sessions lasting 2 and 6 hours both account for 7 respondents. Additionally, 3 and 8 hours of study time were chosen by 6 respondents each. Meanwhile, only 5 respondents reported studying for 1 hour, and just 1 respondent studied for 9 hours. This indicates that the analysis result primarily reflects students who study for moderate durations, around 4 to 7 hours.

3.1.9 Consecutive Study Hours Before Break Distribution

Study Hours	Frequency	Relative Frequency	Cumulative Frequency (CF)	Percent (%) of CF
1	7	0.109	7	10.94
2	21	0.328	28	43.75
3	21	0.328	49	76.56
4	12	0.188	61	95.31
5	3	0.047	64	100
Total	**64**	**1.000**	**64**	**100.00**

Table 3.6 Frequency Distribution Table for Consecutive Study Hours Before Break Distribution

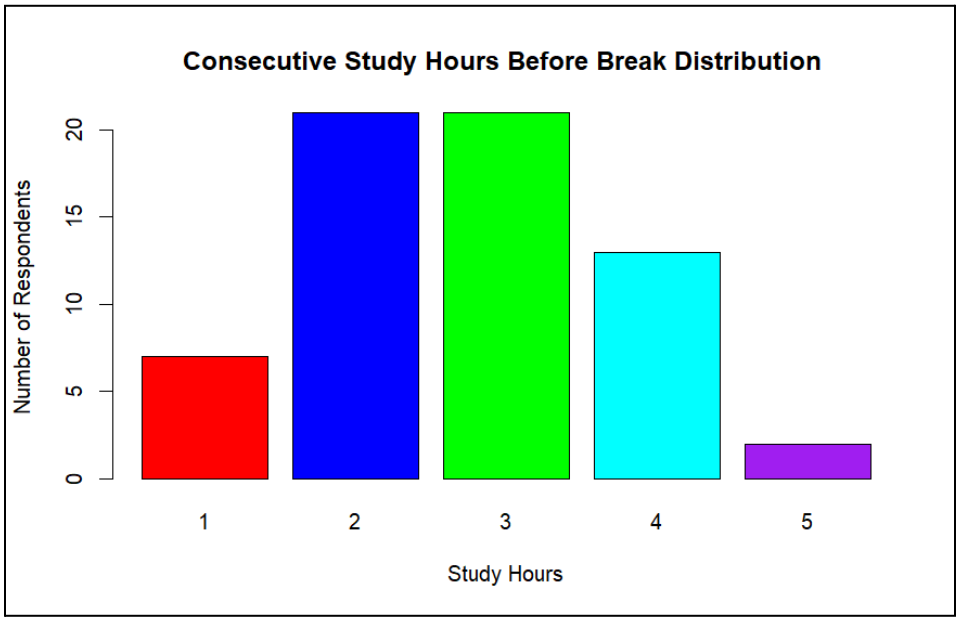


Figure 3.9 Bar Chart for Consecutive Study Hours Before Break Distribution

Based on the bar chart and frequency table, it is shown that students most commonly study for 2-3 hours consecutively before taking a break. The next most common pattern is studying for 4 hours consecutively, with 12 respondents (18.8%) reporting this duration. Both 2-hour and 3-hour study sessions are equally common, with 21 respondents (32.8% each) reporting these durations. This means that nearly two-thirds (65.6%) of all students surveyed prefer to study for 2-3 hours before taking a break.

3.1.10 Break Duration Before Next Revision Distribution

Break Duration (Hours)	Frequency	Relative Frequency	Cumulative Frequency (CF)	Percent (%) of CF
1	36	0.563	36	56.25
2	19	0.297	55	85.94
3	5	0.078	60	93.75
4	2	0.031	62	96.88
5	2	0.031	64	100
Total	**64**	**1.000**	**64**	**100.00**

Table 3.7 Frequency Distribution Table for Break Duration Before Next Revision Distribution

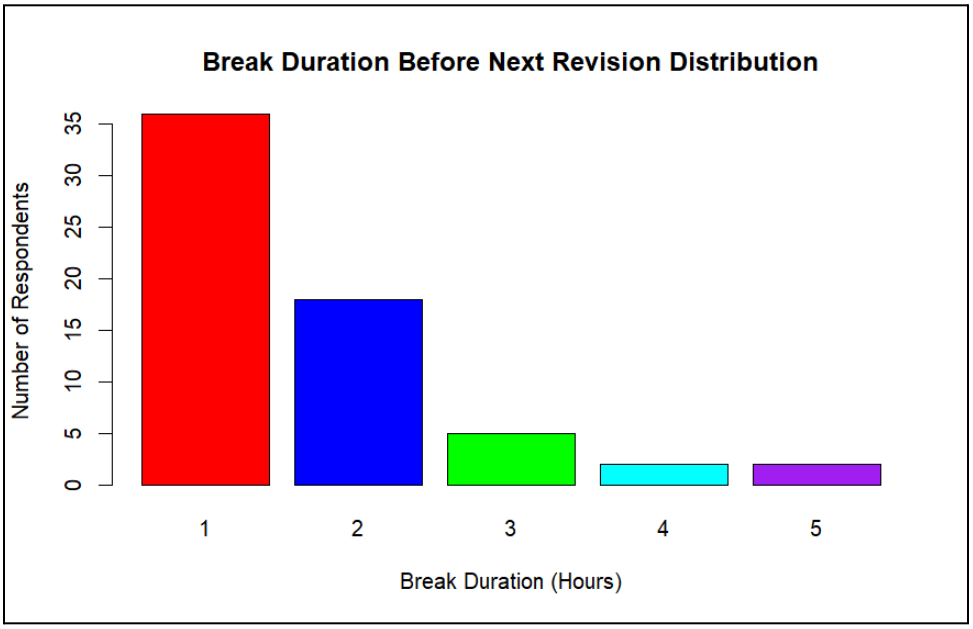


Figure 3.10 Bar Chart for Break Duration Before Next Revision Distribution

Based on the bar chart and frequency table, it is shown that the majority of students take relatively short breaks before returning to their studies. The data reveals that 1-hour breaks are overwhelmingly preferred, with 36 respondents (56.3%) reporting this duration. This means more than half of all surveyed students only take a 1-hour break before resuming their revision. The second most common break duration is 2 hours, reported by 19 respondents (29.7%). Together, these shorter break durations (1-2 hours) account for 86% of all responses, indicating that most students prefer to keep their breaks relatively brief.

3.1.11 Days In Advance To Start Studying Before Exam Distribution

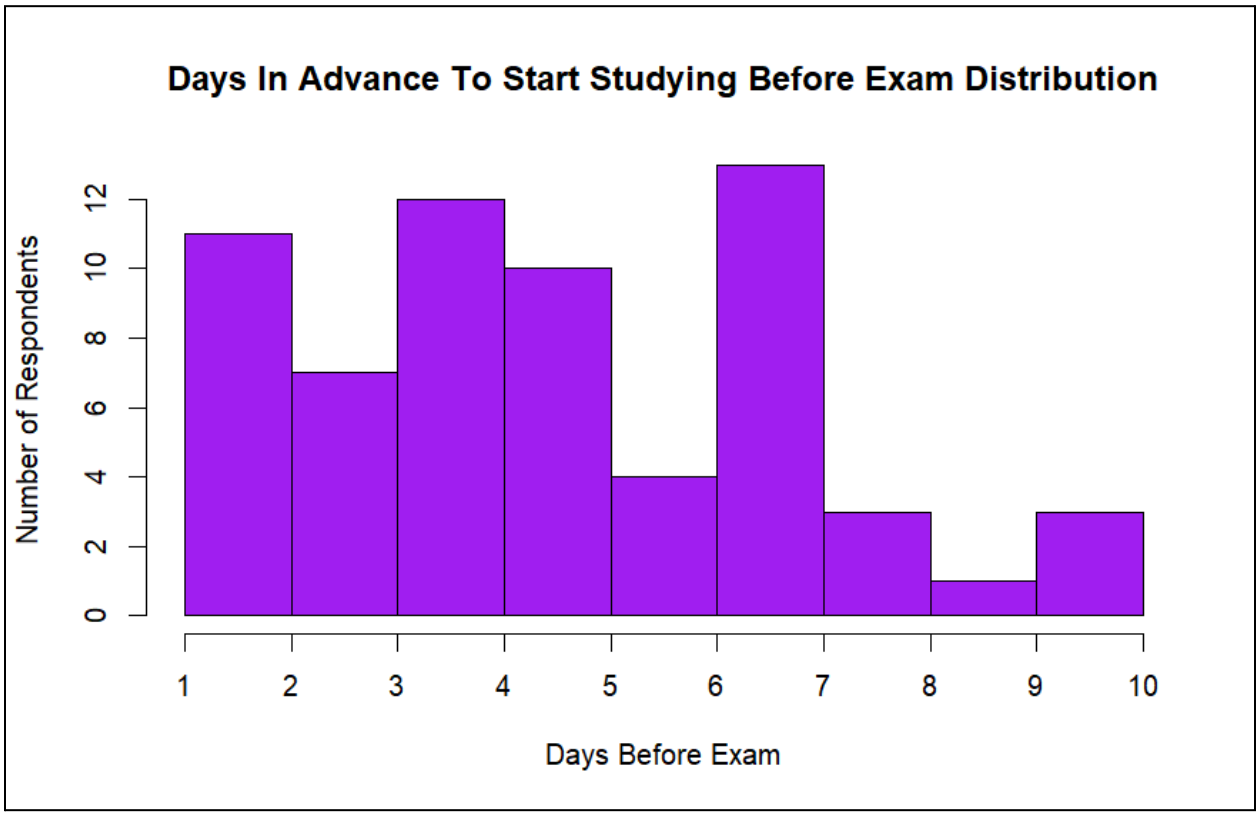


Figure 3.11 Histogram for Days in Advance To Start Studying Before Exam Distribution

Based on the histogram, the data reveals that the most common preparation time is 7 days before an exam, with approximately 13 respondents choosing to begin their studies a week ahead. This represents the highest peak in the distribution. The second most frequent starting points are around 3-4 days before exams, with about 12 respondents starting 4 days ahead and 10 respondents starting 5 days before. There's also a notable group of students (about 11 respondents) who begin studying just 2 days before their exams. This histogram suggests that while the most common approach is to begin studying a week before exams, student preparation habits vary considerably, with meaningful numbers starting at various points between 2-7 days before their exams.

3.1.12 Effectiveness of Study Techniques Distribution

Effectiveness	Frequency	Relative Frequency	Cumulative Frequency (CF)	Percent (%) of CF
1	0	0	0	0
2	0	0	0	0
3	15	0.234	15	23.44
4	22	0.343	37	57.81
5	27	0.421	64	100.00
Total	**64**	**1.000**	**64**	**100.00**

Table 3.8 Frequency Distribution Table for Effectiveness of Study Techniques Distribution

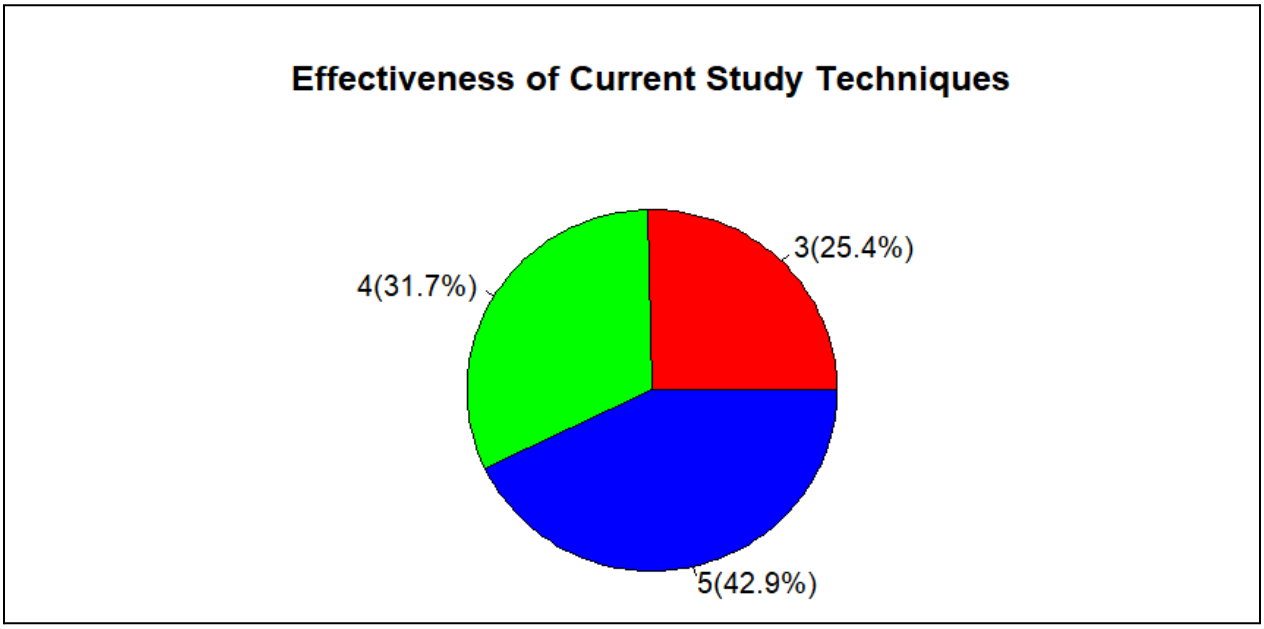


Figure 3.12 Pie Chart for Effectiveness of Study Techniques Distribution

Based on the chart, most students consider their current study techniques to be effective. The largest group (42.9%) rated their study methods as highly effective (5), while 31.7% rated them as effective (4), and 25.4% gave a moderate effectiveness rating (3). In total, about three-quarters of students (74.6%) rated their study techniques as either effective or highly effective. No students rated their study techniques below a 3, suggesting that all students believe their methods have at least moderate effectiveness.

3.2 Possible Analysis 2

Possible Analysis 2: Comparison of Study Environment vs CGPA

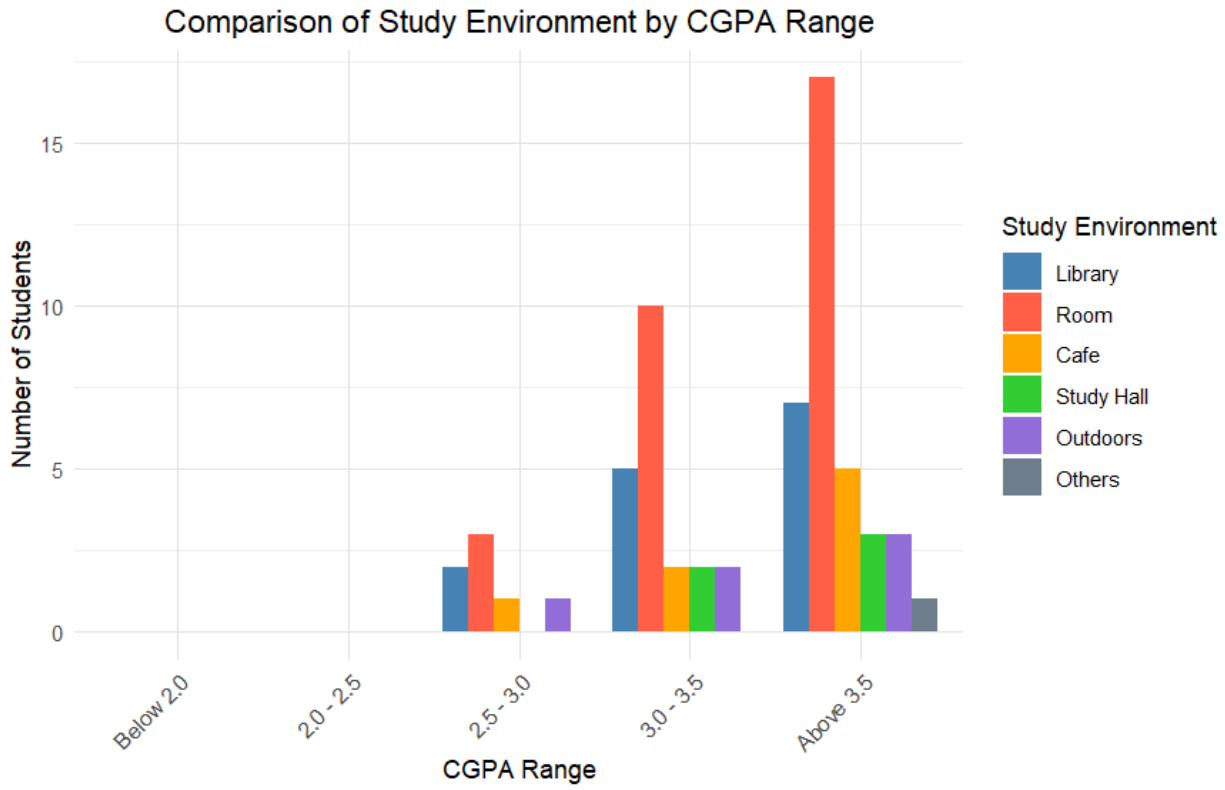


Figure 3.13 Stacked Bar Chart for Comparison of Environment vs CGPA

Based on the bar chart, the data shows that no students reported CGPAs below 2.5, while across all visible CGPA ranges, the Room environment is consistently the most popular choice, with its preference significantly increasing as CGPA rises (from about 3 students in the 2.5-3.0 range to approximately 17 students in the Above 3.5 range). Library consistently ranks as the second most preferred environment across all CGPA groups. Notably, higher-achieving students (Above 3.5 CGPA) demonstrate more diverse study environment preferences, utilizing all six available options (Room, Library, Cafe, Study Hall, Outdoors, and Others), while lower CGPA ranges show fewer environmental preferences. This suggests that higher-performing students may benefit from utilizing a variety of study spaces, though the strong preference for Room environments across all academic performance levels indicates that private, familiar spaces remain most conducive to studying regardless of academic standing.

3.3 Possible Analysis 3

Possible Analysis 3: Comparison of Study Environment vs Level of Study

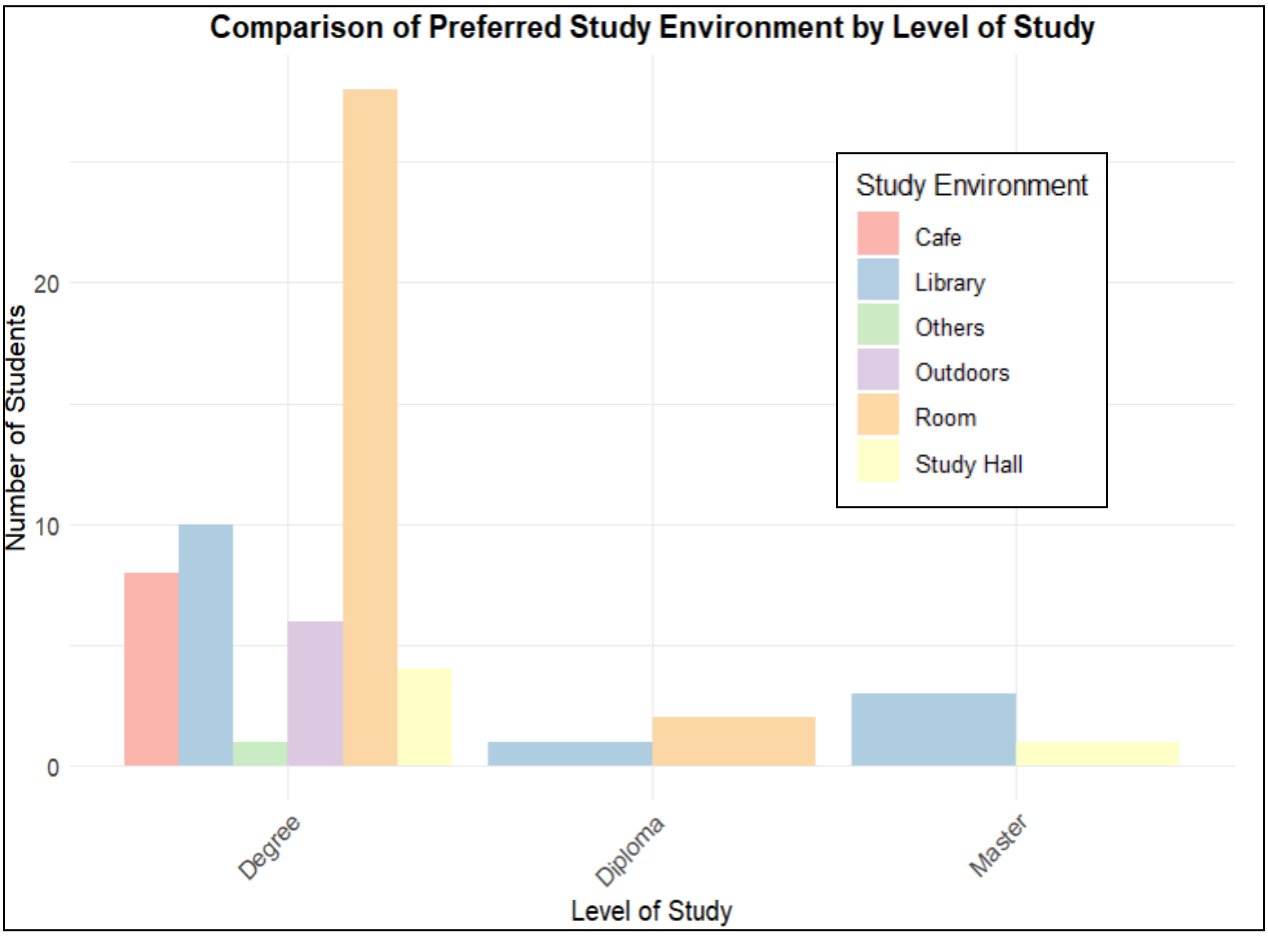


Figure 3.14 Stacked Bar Chart for Comparison of Environment vs Level of Study Distribution

Based on the data, it is shown that the majority of respondents, who are primarily bachelor's degree students, prefer studying in their rooms, with 30 out of 64 respondents selecting this environment. The library is the second most popular study location, chosen by 14 students. Other study environments, such as cafés (8 respondents), outdoors (6 respondents), study halls (5 respondents), and others (1 respondent), are less commonly selected. Given that most respondents are bachelor's degree students (approximately 85.9%), this analysis primarily reflects the study environment preferences of students at the bachelor's level.

3.4 Possible Analysis 4

Possible Analysis 4: Comparison of CGPA vs Study techniques

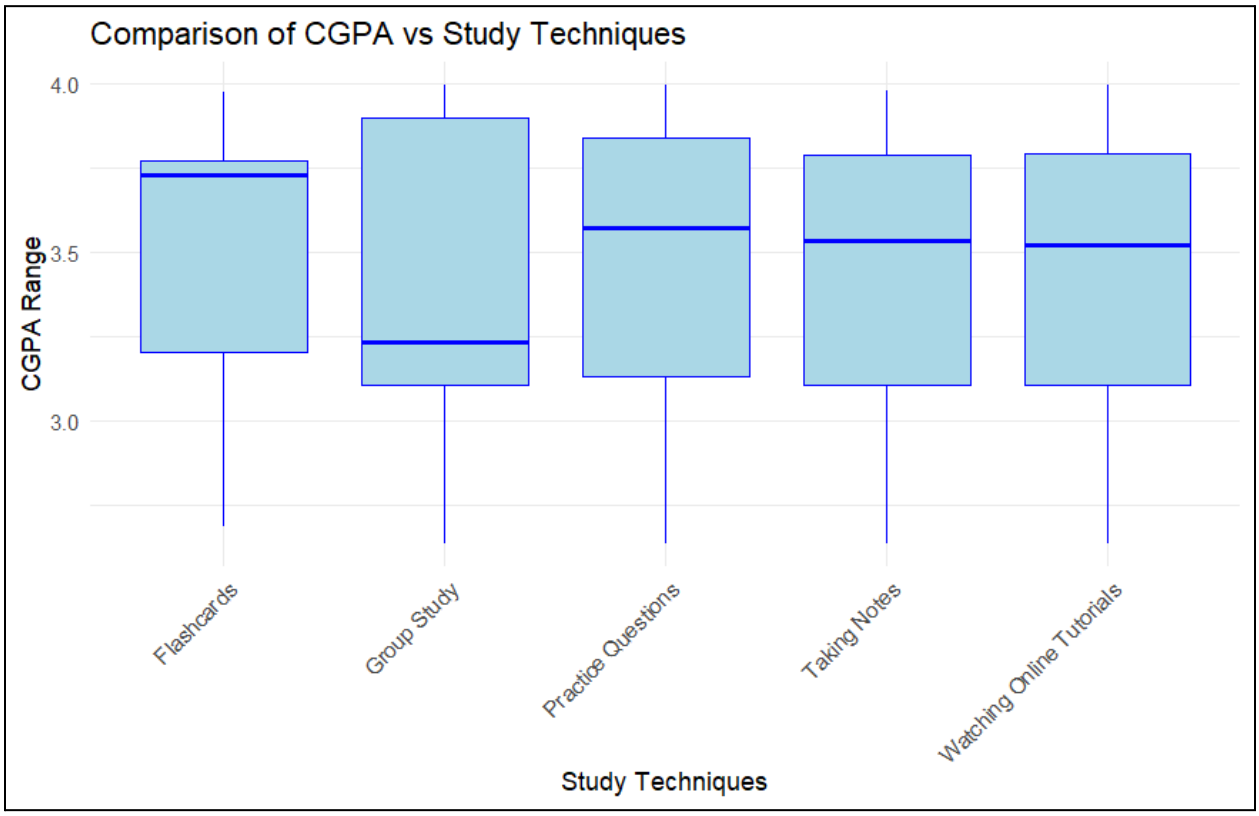


Figure 3.15 Boxplot for Comparison of CGPA vs Study Techniques Distribution

Based on the boxplot, it reveals that Group study appears to have the widest range of CGPA outcomes, with some students achieving higher CGPAs (close to 4.0) while others have lower performance (around 3.0). Practice Questions, Textbooks, and Watching Online Tutorials show similar median CGPA values (around 3.5).

Flashcards show a narrower interquartile range with a slightly higher median CGPA than most other techniques. All five techniques have maximum CGPA values reaching close to 4.0, indicating that top performers use various study methods.

The overlapping ranges across all techniques suggest that no single study method definitively leads to better academic performance than others. Students achieve similar CGPA ranges regardless of their preferred study technique, though there are slight variations in the distribution patterns.

3.5 Possible Analysis 5

Possible Analysis 5: Comparison of CGPA vs Average Revision Time

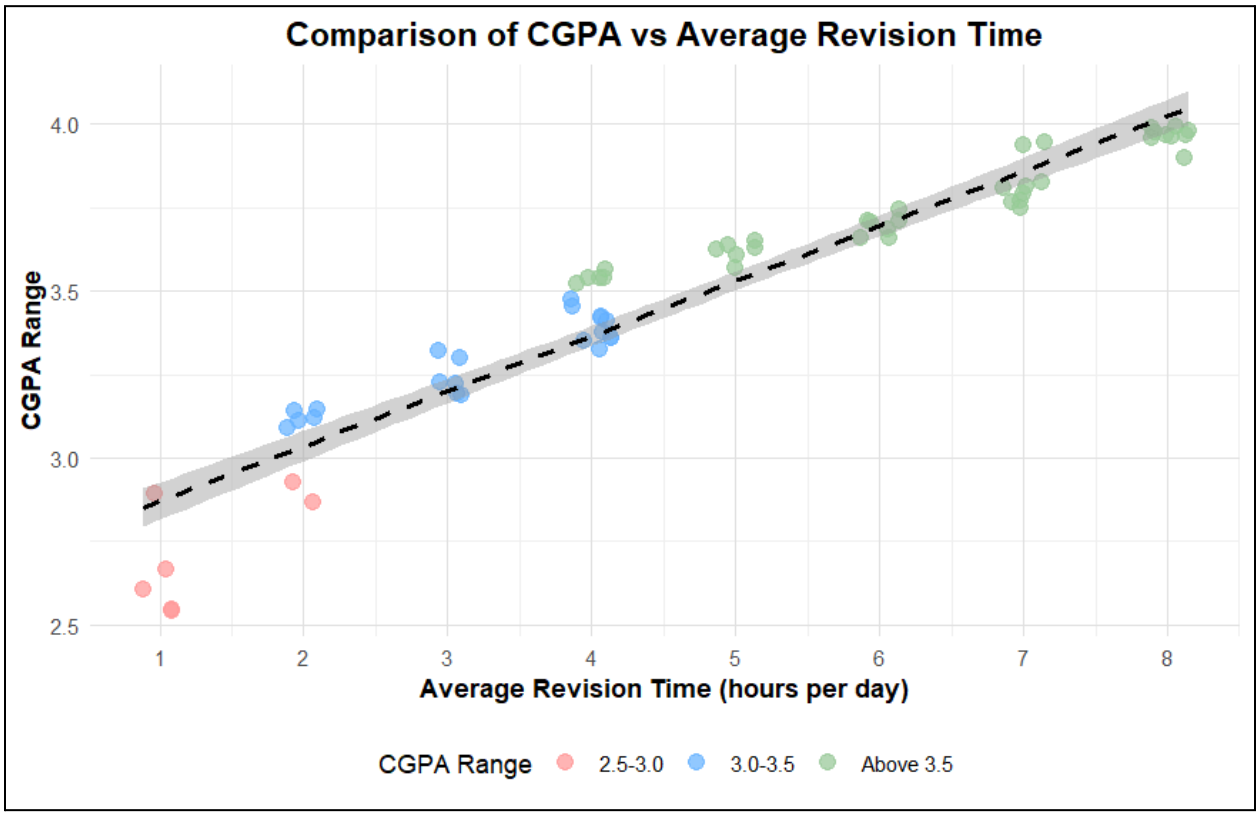


Figure 3.16 Scatter Plot for Comparison of CGPA vs Average Revision Time Distribution

Based on the scatter plot, it reveals a strong positive correlation between average revision time (hours per day) and CGPA outcomes. Students who spend more time on revision (7-8 hours per day) consistently achieve higher CGPAs (above 3.5, approaching 4.0), while those who spend minimal time (1-2 hours per day) tend to have lower CGPAs (2.5-3.0).

The data points are color-coded into three CGPA ranges: 2.5-3.0 (pink), 3.0-3.5 (blue), and above 3.5 (green). Students in the middle range (3-5 hours of revision per day) show a mix of performance outcomes (mostly 3.0-3.5 CGPA), suggesting this is a transitional zone where other factors might influence academic success beyond just time spent.

The tight confidence interval around the trend line suggests the relationship between revision time and CGPA is consistent and reliable across the dataset, with relatively low variability.

3.6 Possible Analysis 6

Possible Analysis 6: Comparison of consecutive hours can students study vs duration of break before next revision session

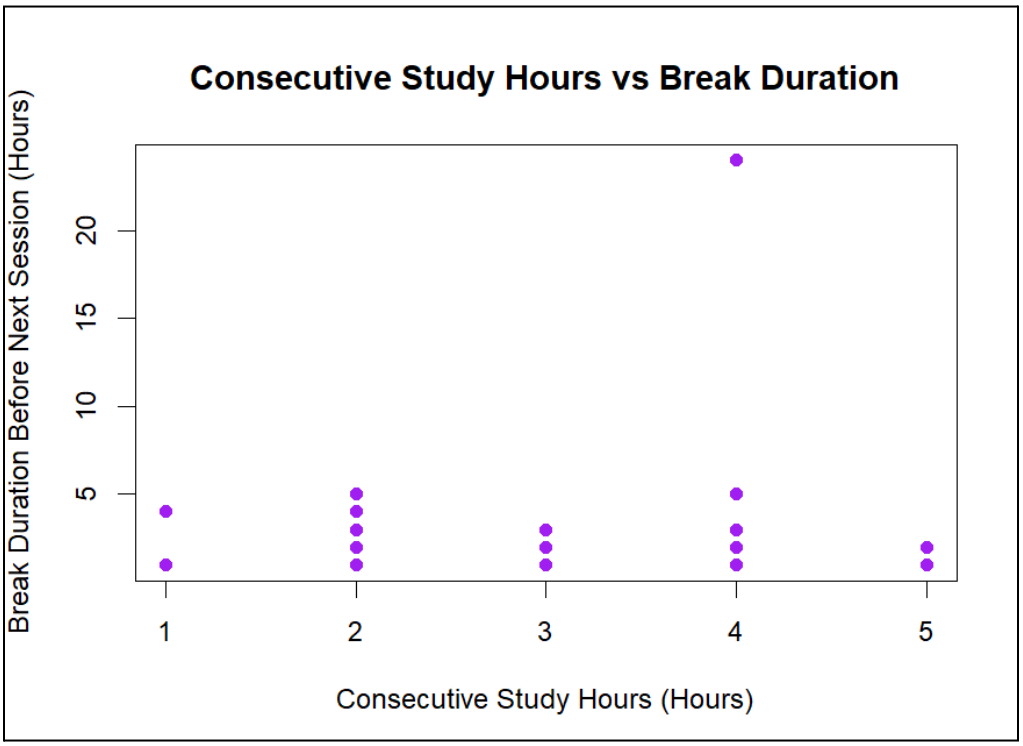


Figure 3.17 Scatter Plot for Comparison Study Hours and Break Duration

Based on the scatter plot above, it is shown that there is no significant relationship between how long students study continuously and how long they break. It also shows that students tend to study 1 to 5 hours consecutively and take 1 to 5 hours break. From the plot, it cannot be seen that longer study hours lead to long break hours. Students seem to have similar length breaks regardless of how long they have studied. For example, students who study for 1, 2, 3, 4, 5 consecutive hours tend to have 1 hour break.

However, it is obvious that there is an outlier in the plot. At 4 consecutive study hours, a student takes about 24 hours, which is a very long break before their next session. This could show that the student studies for 4 hours and then does not return to studying until the next day.

3.7 Possible Analysis 7

Possible Analysis 7: Comparison of CGPA vs. days in advance students begin studying for exams

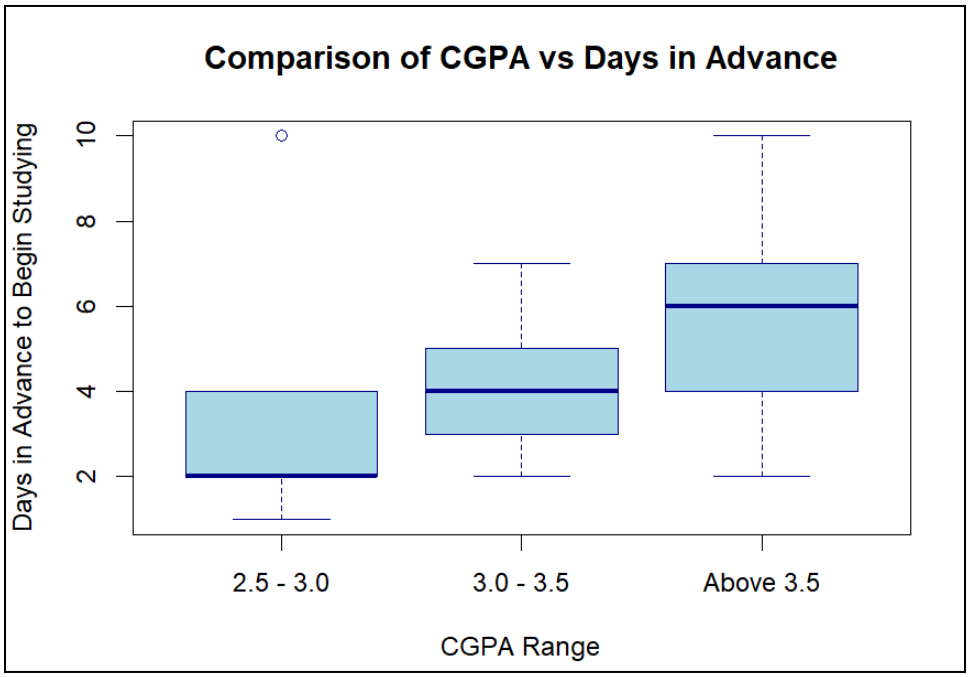


Figure 3.18 Box Plot for Comparison CGPA and Days in Advance

Based on the boxplot above, a clear positive relationship exists between CGPA range and how many days in advance students begin studying. It is shown that the higher the CGPA range, the more days in advance students begin studying. In other words, students with higher CGPA tend to start studying earlier.

In the 2.5-3.0 CGPA range, 2 is the median, meaning that students overall start studying about 2 days in advance. Some students also start studying as late as 1 day before; some start up to 4 days ahead. There is also an outlier, which means that a student starts studying around 10 days in advance, which is uncommon in this group. In the 3.0-3.5 CGPA range, 4 is the median, showing that students in this group typically start studying 4 days in advance. Overall, students start studying between 3 to 5 days before the exam starts.

Lastly, students that are above 3.5 CGPA tend to start studying earlier, with a median of approximately 6 days, and most falling between 4-7 days. In general, students in each higher CGPA range tend to prepare about 2 days earlier.

4.0 Conclusion

Based on the data presented in this study on common study habits among UTM students, multiple key factors regarding effective academic practices and their relationship to result were found. This survey was conducted primarily among bachelor's degree computing students with balanced gender distribution, which reveals important data about study habits. Higher-performing students (with CGPAs above 3.5) typically begin exam preparation 6-7 days in advance and dedicate more daily hours (7-8) to studying compared to their lower-achieving peers. This demonstrates a clear correlation between time investment and academic success, suggesting that early preparation and consistent study are crucial factors. Besides that, most students prefer study sessions of 2-3 consecutive hours followed by brief 1-2 hour breaks, with a strong majority rating their established study methods as effective. This indicates that students have developed sustainable study patterns that work well with their cognitive rhythms and learning styles. While various study techniques (note-taking, practice questions, watching tutorials, flashcards, group study) are widely used, no single method demonstrated dramatically superior results compared to others. This suggests that consistency and time invested may be more important factors in academic success than the specific techniques employed. These findings indicate that academic achievement at UTM is closely tied to disciplined study habits, particularly regarding preparation time and consistency. The insights from this research could help guide students in developing more effective study strategies to improve their academic performance. Also, through this study, statistical analysis skills and data interpretation skills of our team have significantly improved. We learn how to use R programming to do descriptive statistics to summarize demographic characteristics of the sample.

5.0 References

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5.0 Appendix

Google Form: https://shorturl.at/0tWrF

Google Sheet: https://docs.google.com/spreadsheets/d/1GcTq7FmwxTPfFRMedNYGwFKLDZ1PNmasSTwKAGyjEuk/edit?usp=sharing